Nonuniqueness for a BSDE with generator $|z| \sin |z|$

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Abstract

We construct two solutions with the same terminal value for the scalar Brownian BSDE with generator $g(z) = |z| \sin |z|$. Both solutions have integrable supremum powers of every finite order. This gives a negative answer to Fan, Hu and Tang's Problem 5.3.

Fix $T > 0$ and the augmented natural filtration of a one-dimensional standard Brownian motion B. Fan, Hu and Tang ask whether

$$Y_t = \xi + \int_t^T |Z_s| \sin |Z_s| \, ds - \int_t^T Z_s \, dB_s \tag{1}$$

has a unique solution when $\xi \in L^p$, $p > 1$, and $(|Y_t|^p)$ is of class (D) [1, Problem 5.3, equation (5.2)]. Here class (D) means that the stopped variables $|Y_\tau|^p$, $\tau \le T$, form a uniformly integrable family.

Theorem 1. *For every $T > 0$, there are an $\mathcal{F}_T$-measurable $\xi \in \bigcap_{1 \le p < \infty} L^p$ and two solutions (Y^+, Z^+), (Y^-, Z^-) of (1) such that, for every finite $p \ge 1$,*

$$\mathbb{E} \sup_{t \le T} |Y_t^{\pm}|^p + \mathbb{E} \left(\int_0^T |Z_t^{\pm}|^2 \, dt \right)^{p/2} < \infty.$$

Moreover, $Y_t^+ > Y_t^-$ for every $t < T$, almost surely.

Proof. We use Itô's formula, stopped martingale identities and the Burkholder–Davis–Gundy inequality in their usual continuous-martingale forms; see [2, Chapters IV and IX].

1. A positive process with an integrable inverse-square clock. Let

$$dR_t = dB_t + R_t^{-1} \, dt, \qquad R_0 = 1, \qquad A_t = \int_0^t R_s^{-2} \, ds. \tag{2}$$

The stopped Picard construction gives a strong solution up to exit from $(0, \infty)$, since the coefficients are locally Lipschitz there. For $n \ge 2$, let σ_n be its first exit from $(1/n, n)$ and put $\tau_n = T \wedge \sigma_n$. Itô's formula gives

$$d(R_t^{-1}) = -R_t^{-2} \, dB_t, \qquad d(R_t^2) = 2R_t \, dB_t + 3 \, dt.$$

The stopped integrands are bounded, so $\mathbb{E} R_{\tau_n}^{-1} = 1$ and $\mathbb{E} R_{\tau_n}^2 = 1 + 3\mathbb{E}\tau_n \le 1 + 3T$. Consequently

$$\mathbb{P}(\sigma_n \le T) \le \frac{1}{n} + \frac{1 + 3T}{n^2}.$$

Thus R exists throughout $[0,T]$ and stays strictly positive.

Another application of Itô's formula shows that

$$L_t = R_t^{-1/2}\exp(A_t/8), \qquad dL_t = -\frac{L_t}{2R_t}\,dB_t.$$

In particular $\mathbb{E}L_{\tau_n} = 1$. Cauchy–Schwarz and the preceding second-moment bound yield

$$\mathbb{E}e^{A_{\tau_n}/16} = \mathbb{E}\big[L_{\tau_n}^{1/2}R_{\tau_n}^{1/4}\big] \le (\mathbb{E}R_{\tau_n}^{1/2})^{1/2} \le (1+3T)^{1/8}.$$

Fatou's lemma therefore gives

$$\mathbb{E}e^{A_T/16} \le (1+3T)^{1/8} < \infty. \tag{3}$$

In particular A_T has moments of every finite order.

2. A nonzero gap that vanishes at the terminal time. Let φ and Φ denote the standard normal density and distribution function. For $0 \le h \le T$, $r > 0$, define

$$u(h,r) = \frac{1}{2\sqrt{T}}\int_0^h \Phi(-r/\sqrt{s})\,ds, \qquad u(0,r) = 0. \tag{4}$$

Differentiating for $h, r > 0$ gives

$$u_h = \frac{1}{2\sqrt{T}}\Phi(-r/\sqrt{h}), \qquad u_{rr} = \frac{1}{\sqrt{T}}\Phi(-r/\sqrt{h}), \qquad u_r = -\frac{1}{2\sqrt{T}}\int_0^h \frac{\varphi(r/\sqrt{s})}{\sqrt{s}}\,ds.$$

For the second identity, integrate $\partial_s\Phi(-r/\sqrt{s}) = r\varphi(r/\sqrt{s})/(2s^{3/2})$. These differentiations are justified by exponential decay as $s \downarrow 0$. Thus $u_h = \frac{1}{2}u_{rr}$, $-1 \le u_r \le 0$, and $0 \le u \le \sqrt{T}/4$; the derivative bound follows from $\varphi \le 1$ and $h \le T$.

Set

$$D_t = u(T-t, R_t), \qquad q_t = u_r(T-t, R_t), \qquad q_T = 0.$$

The processes are continuous, $D_T = 0$, and $D_t > 0$ for $t < T$. Using (2) and the heat equation gives

$$dD_t = \frac{q_t}{R_t}\,dt + q_t\,dB_t. \tag{5}$$

The identity extends to T by continuity: q is bounded and every path of R has a positive minimum on $[0,T]$.

3. Realizing the required secant of the generator. For $q \in [-1,0]$, write

$$s(q) = \begin{cases} \sin(q/2)/(q/2), & q \ne 0, \\ 1, & q = 0, \end{cases} \qquad s_0 = 2\sin(1/2).$$

Then $0 < s_0 \le s(q) \le 1$ and $\cos(q/2) > 0$. For $r > 0$ put $k(r) = \lceil (2\pi s_0 r)^{-1} \rceil$. On the interval

$$I_r = [2\pi k(r) + \pi/2,\ (2k(r)+1)\pi]$$

the function

$$F(a,q) = \sin a\cos(q/2) + a\cos a\, s(q)$$

is strictly decreasing, because

$$F_a = (\cos(q/2) + s(q))\cos a - a\, s(q)\sin a < 0.$$

Its left endpoint value is $\cos(q/2) > 0$, and its right endpoint value is at most $-1/r$. There is therefore a unique $a = a(r,q) \in I_r$ with $F(a,q) = -1/r$. This choice is Borel: on each Borel set where $k(r)$ is constant, the strictly monotone inverse depends continuously on (r,q). It satisfies

$$0 < a(r,q) \le \frac{1}{s_0 r} + 3\pi. \tag{6}$$

Both $a + q/2$ and $a - q/2$ are positive. The addition formula for sine now gives the exact identity

$$g(a + q/2) - g(a - q/2) = qF(a,q) = -q/r. \tag{7}$$

The formula also holds at $q = 0$.

Define predictable processes

$$Z_t^{\pm} = a(R_t, q_t) \pm q_t/2, \qquad Y_t^- = -\int_0^t g(Z_s^-)\, ds + \int_0^t Z_s^-\, dB_s, \qquad Y_t^+ = Y_t^- + D_t.$$

By (6), $|Z_t^{\pm}| \le C(1 + R_t^{-1})$ for a fixed constant C. Hence (3) implies

$$\mathbb{E}\left(\int_0^T |Z_t^{\pm}|^2\, dt\right)^{p/2} < \infty \qquad (1 \le p < \infty).$$

Since $|g(z)| \le |z|$, Cauchy–Schwarz controls the drift integral, and the Burkholder–Davis–Gundy inequality controls the stochastic integral. Thus $\mathbb{E}\sup_{t\le T}|Y_t^-|^p < \infty$ for every finite $p \ge 1$. The same holds for Y^+ because D is bounded.

Finally, (5) and (7) give

$$dY_t^+ = -g(Z_t^-)\, dt + Z_t^-\, dB_t + \frac{q_t}{R_t}\, dt + q_t\, dB_t = -g(Z_t^+)\, dt + Z_t^+\, dB_t.$$

Taking $\xi = Y_T^- = Y_T^+$ proves (1) for both pairs. Their initial values differ by $u(T,1) > 0$. For each $p \ge 1$, the integrable variable $\sup_{t\le T}|Y_t^{\pm}|^p$ dominates all stopped powers, establishing the required class-(D) condition. □

The same counterexample works for any Brownian dimension $d \ge 1$: use its first coordinate for B and replace each scalar $Z^{\pm}$ by $(Z^{\pm}, 0, \ldots, 0)$.

Use of AI tools

The author used research AI agents that the author built and configured (GPT-6 Astra max, Claude Opus 5.5 max and GPT-6 Astra Pro) during the proof work, for proof search and construction, proof checking, translation, Python computation and similar tasks. The final manuscript was written, and the proofs were checked, by the human author. All responsibility for this work rests with the author personally.

References

[1] S. Fan, Y. Hu and S. Tang, *1D nonlinear backward stochastic differential equations: a unified theory and applications*, arXiv:2309.06233v2, 11 February 2026. Problem 5.3 and equation (5.2). https://arxiv.org/html/2309.06233v2#S5.

[2] D. Revuz and M. Yor, *Continuous Martingales and Brownian Motion*, 3rd ed., Springer, 1999. Chapters IV and IX. https://doi.org/10.1007/978-3-662-06400-9.